Vaibhav Sethi

APPLIED STATISTICS · MACHINE LEARNING · QUANTITATIVE & CREDIT RISK

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SUMMARY

Fourth-year UBC B.Sc. student (Statistics & Physics) targeting **data-analyst** and **quantitative / credit-risk analyst** roles. Built a **CCAR/ DFAST-style mortgage stress-testing engine** on 350K+ real Freddie Mac loans and a calibrated credit-scoring platform with SHAP explainability and expected-loss simulation. Fluent across **Python, SQL, R, and SAS**; validates models the way regulators expect — Gini/ KS/PR-AUC, probability calibration, out-of-time testing, and model cards. Bloomberg Market Concepts certified. Coursework complete Dec 2026; available for new-grad and analyst roles from early 2027 (open to Fall 2026 co-op).

EDUCATION

University of British Columbia — *B.Sc., Combined Major in Statistics and Physics* Coursework Dec 2026 · Convocation May 2027

Relevant coursework: Regression & GLMs, Time Series, Multivariate Analysis, PCA & Clustering, Statistical Inference, Machine Learning, Experimental Design, Statistical Computing (R & Python).

TECHNICAL SKILLS

Languages **Python** (pandas, NumPy, scikit-learn, XGBoost, statsmodels, SHAP) · **SQL** (DuckDB — window functions, CTEs) · **R** (tidyverse, ggplot2, Shiny, glm) · **SAS** (PROC LOGISTIC, PROC SQL) · Embedded C

Statistics/ML Logistic & Linear Regression · GLMs · Discrete-time Survival/Hazard Models · Time Series · PCA & Clustering · Probability Calibration (Platt, isotonic) · Cross-Validation · Imbalanced Classification

Credit Risk PD / LGD / EAD & Expected Loss · CCAR / DFAST-style Stress Testing · ROC / PR-AUC / Gini / KS · SHAP Reason Codes · Model Cards & Governance (SR 11-7)

Data & BI DuckDB (tens-of-millions-of-row pipelines) · Parquet · **Power BI** (star schema) · **Excel** (financial modeling) · Tableau · Git & GitHub Pages · Jupyter · RStudio · LaTeX · Bloomberg Terminal (BMC)

PROJECTS

Mortgage Stress-Testing Engine — Credit Risk under Fed 2026 Scenarios

Python · SQL · SAS · R · DuckDB · XGBoost · SHAP · Power BI · Excel

- Built a loan-level **probability-of-default** model on **350K+ real Freddie Mac mortgages**, engineering features across **19M loan-month records in DuckDB**; validated **out-of-time by vintage** at **Gini 0.66 / KS 0.50 / PR-AUC 0.34** and calibrated with isotonic regression (Brier 0.175→0.064), with SHAP reason codes.
- Fit a **discrete-time hazard model with macroeconomic covariates** and projected portfolio Expected Loss over a 13-quarter horizon under the Federal Reserve's 2026 baseline vs severely-adverse **(CCAR/DFAST)** scenarios — severely-adverse losses ran **5.5× baseline** (0.99% vs 0.18%; $352M vs $65M on a $35B book).
- Shipped an interactive **web dashboard + stress simulator** and a model card for governance; reproduced the PD model in **SAS (PROC LOGISTIC)** and **R (glm)**, plus a **Power BI** star-schema export.

Banking Risk Platform — Credit Scoring & Expected Loss

R · XGBoost · Shiny · SHAP · tidyverse

- Built an end-to-end credit-scoring platform in R: **XGBoost PD model** benchmarked vs logistic regression under severe class imbalance with stratified k-fold CV; evaluated on **ROC AUC and PR-AUC** (not accuracy) and calibrated with Platt scaling for usable PD estimates.
- Simulated portfolio **expected loss (PD × LGD × EAD)**, delivered per-prediction **SHAP reason codes** and a model card for governance, and shipped a **Shiny dashboard** for single & batch CSV scoring.

LIGO Gravitational-Wave Glitch Classifier

Python · scikit-learn · Random Forest · Grid Search

- Built a multi-class classifier on **7,966 detector-noise samples across 22 severely imbalanced classes** (67× ratio); tuned Random Forest via grid search and 3-fold CV → **88% accuracy, 0.81 macro F1**, with CV and test within 0.01 (no overfit).

Vancouver Housing Price Prediction

R · Python · ggplot2 · Time Series

- Modeled residential prices using **10+ years of transaction data** joined with macro indicators (rates, CPI, income); engineered spatial/ temporal/property features and validated on held-out sets via RMSE/MAE.

EXPERIENCE

Staples Canada — *Printing Services Representative* Dec 2022 – Jun 2023 · Vancouver, BC

- Processed **50–70 client print orders per shift** on tight deadlines while keeping quality consistent; resolved equipment failures under pressure through systematic troubleshooting.

CERTIFICATIONS & ACTIVITIES

Bloomberg Market Concepts (BMC) · UBC Data Science Club, Member (2025–Present)